Auteur: Michael Livchits
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$$w = \ln(u + v + z) \text{ met } u = \cos^2 t,\ v = \sin^2 t \text{ en } z = t^2$$

$$\frac{\partial w}{\partial u} = \frac{1}{u + v + z}$$

$$\frac{\partial w}{\partial v} = \frac{1}{u + v + z}$$

$$\frac{\partial w}{\partial z} = \frac{1}{u + v + z}$$

$$\frac{\partial u}{\partial t} = -2\cos t \sin t$$

$$\frac{\partial v}{\partial t} = 2\sin t \cos t$$

$$\frac{\partial z}{\partial t} = 2t$$

$$Dw(u, v, w) = \begin{pmatrix} \frac{1}{u+v+z} & \frac{1}{u+v+z} & \frac{1}{u+v+z} \end{pmatrix}$$

$$D(u, v, z)(t) = \begin{pmatrix} -2\cos t \sin t \\ 2\sin t \cos t \\ 2t \end{pmatrix}$$

$$Dw(u, v, z) \cdot D(u, v, z)(t) = \frac{2t}{u + v + z} = \frac{2t}{1 + t^2}$$

References